



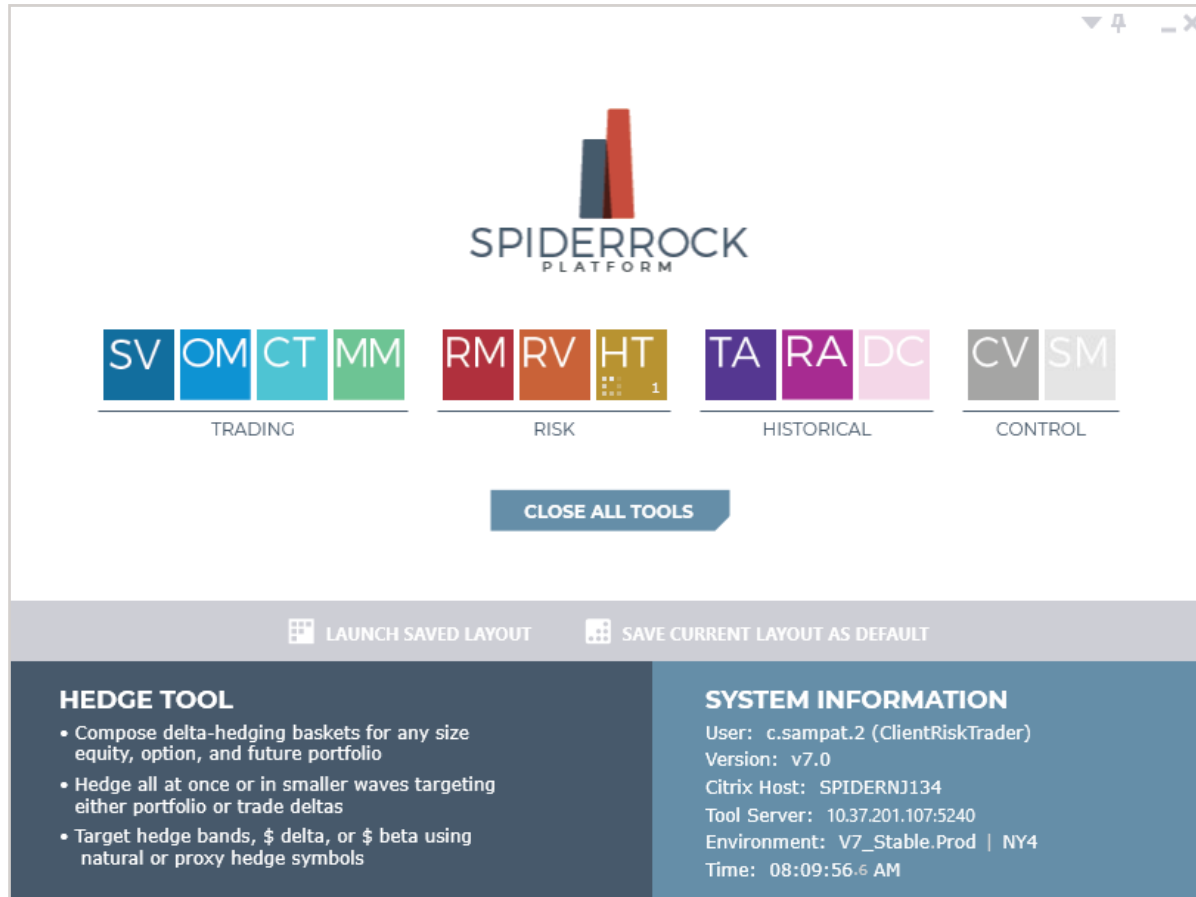
SpiderRock V7 Tool Concept Guide

Hedge Tool

March 16, 2021

OVERVIEW

Versatile tool allowing customers to dynamically hedge any portfolio or sections of a portfolio. Users can hedge the delta or beta of their current positions, open positions or day trades using natural or proxy symbols. Users can also mitigate the cost of hedging by setting a risk tolerance limiting the hedging to the nearest delta band, and by choosing a mid market execution algo. Users can choose to hedge the portfolio all at once or send subsequent waves to automatically decrement the delta exposure to the desired level over a user defined period of time.



HEDGE TOOL FUNCTIONALITIES

The Hedge Tool is comprised of three sections: Hedging Parameters, Totals Boxes, and Current Position Detail. Each section is summarized below and described in more detail on the following pages.

- Hedging Parameters allow users to set hedge target, index allocation, bandwidth (risk tolerance), portfolio size, max order size/wave and execution algorithms.
- Totals Boxes summarize the current position risk associated with the portfolio or the portion of portfolio displayed in the Current Position Detail window. It also indicates how much dollar delta is being bought or sold (green boxes).
- Current Position Detail displays risk metrics associated to start of day positions plus any day trades at the root symbol level.

SPIDERROCK Hedge Tool

Quick filter: All

Symbol(s): <ALL>

All

View: Regular

Test Account: T.CH3

Refresh

Hedge Once

Start Loop

Cancel Wave

Hedge Target

Current Positions

Delta

Index Allocation

Comp to Target: Uniform

\$delta cap: 100,000,000

Bandwidth

hRatio: 0.25

\$delta: 0

z5Mult: 99

Portfolio

100 %

Reset

Max Order Size / Wave

shr: 500,000

\$de: 50,000,000

#Syms

pos\$delta

pos\$beta

opn\$delta

opn\$beta

shBot

shSld

bHdg\$De

sHdg\$De

Hdg\$De

bOrd\$De

sOrd\$De

Ord\$De

Totals Boxes

Hedging Parameters

23

62.62 m

59.57 m

62.48 m

59.44 m

500

0

6.01 m

67.70 m

(61.68 m)

6.01 m

65.90 m

(59.89 m)

HEDGING PARAMETERS

- Users can hedge the portfolio once or set waves that will work for a user defined time-in-force. This method will decrement the deltas until the nearest risk band is met or the user cancels the wave.

The screenshot displays the Hedging Parameters interface. On the left, there are three buttons: "Hedge Once", "Start Loop", and "Cancel Wave", which are highlighted with a red box. Below these buttons is a table showing the current state of the portfolio. The table has columns for #Syms, pos\$delta, pos\$beta, opn\$delta, opn\$beta, shBot, and shSl. The values are: #Syms: 5, pos\$delta: 8.51 m, pos\$beta: 13.19 m, opn\$delta: 8.51 m, opn\$beta: 13.19 m, shBot: 0, shSl: 0. Below this table is another table with columns: symbol, acctnt, alloc, riskClass, stkStatus, de.src, and opDay. The rows are: 1 VIX T.AB None Hold IV, 2 AMZN T.AB None TwoWay IV, 3 SPX T.AB Component Hold IV, 4 AAPL T.AB None TwoWay IV. On the right, there is a section titled "Execution Algorithm" with a dropdown menu for algoType set to "AUTO.MID" and a time-in-force (tif) dropdown menu set to "10 sec". A red arrow points to the "AUTO.MID" dropdown.

#Syms	pos\$delta	pos\$beta	opn\$delta	opn\$beta	shBot	shSl
5	8.51 m	13.19 m	8.51 m	13.19 m	0	0

	symbol	acctnt	alloc	riskClass	stkStatus	de.src	opDay
1	VIX	T.AB	None		Hold	IV	
2	AMZN	T.AB	None		TwoWay	IV	
3	SPX	T.AB	Component		Hold	IV	
4	AAPL	T.AB	None		TwoWay	IV	

Execution Algorithm
algoType: AUTO.MID tif: 10 sec

dnBand	hedge	orde
-724	22,606	
-215	30	
-94	105	
-1,455	-57,212	
-940	85	72 265.88

- Hedge Once Button:** Users can choose to hedge once. The basket of orders will work for the time-in-force (tif) selected.
- Start Loop:** Users can send waves of orders and each wave will work for the time-in-force selected. This would decrement the deltas in a looping fashion until manually canceled or until hedging is complete.
- Cancel Wave:** This will prompt the system to stop any waves of orders that were previously triggered by the user via the Start Loop button.

Hedge Once		Start Loop		Cancel Wave		Hedge Target	
#Syms 5		pos\$delta 9.74 m		pos\$beta 14.74 m		opn\$delta 8.75 m	
						Open Positions Current Positions Open Positions Trades Trades Protected	
	symbol	acct	alloc				
1	VIX	T.AB	None				
2	AMZN	T.AB	None		TwoWay		
3	SPX	T.AB	Component		Hold		
4	AAPL	T.AB	None		TwoWay		
5	SPY	T.AB	Index		TwoWay		

Cancel Wave		Hedge Target	
		Open Positions Delta Delta Beta	
pos\$beta 4.74 m		opn\$delta 8.75 m	
		opn\$beta 13.75 m	
	alloc	riskClass	stkStatus
	None		Hold
	None		TwoWay
	Component		Hold
	None		TwoWay
	Index		TwoWay

• Hedge Target

Users can set the hedge target to the following options:

- Current Positions: Hedging Start of Day (SOD) positions plus any day trades
- Open Positions: Hedging Start of Day (SOD) positions only
- Trades: Hedging day trades only. SOD positions are not taken into account
- Trades Protected: Hedging day trades only, but will not hedge trades that offset current position risk

Users have a choice between the following options for hedging based on delta or beta:

- Delta: The hedging target is based on \$ deltas
- Beta: The hedging target is based on \$ betas

Important note: The Hedge Tool should be used carefully by users who autohedge any options orders using the Symbol Viewer.

- **Index Allocation**

Users have a choice to hedge a position naturally with the corresponding underlyer instrument (no allocation as displayed in the dropdown list below), or via a proxy instrument (Comp to Target: Uniform/Comp to Target: Day Gm).

The screenshot shows the SPIDERROCK Hedge Tool interface. At the top, there's a 'Quick filter: All' dropdown. Below it are three buttons: 'Hedge Once', 'Start Loop', and 'Cancel Wave'. To the right of these buttons are two dropdown menus: 'Hedge Target' (set to 'Current Positions') and 'Index Allocation' (set to 'No Allocation'). Below the 'Index Allocation' dropdown, a list of options is visible: 'No Allocation', 'Comp to Target: Uniform', and 'Comp to Target: Day Gm'. At the bottom, there are several data fields: '#Syms' (63), 'pos\$delta' (507.81 m), 'pos\$beta' (2,656.77 m), 'opn\$delta' (510.23 m), 'opn\$beta' (2,668.55 m), and 'shBot' (0).

Below is an example of a portfolio that is hedged with the corresponding underlyer instruments. Please note that the default value under the alloc column is "None".

The screenshot shows the SPIDERROCK Hedge Tool interface with a table of portfolio positions. The table has columns for '#Syms', 'symbol', 'acctnt', 'stk.status', 'alloc', 'delta', 'gamma', 'vega', 'pos.de', 'hedge', 'order', and 'pos\$de'. The 'alloc' column is highlighted with a red box, showing 'None' for all positions. The 'hedge' column shows values for each position, and the 'order' column shows values for each position. The 'pos\$de' column shows values for each position. The table is filtered by 'Quick filter: All' and 'Symbol(s): <ALL>'. The 'Hedge Target' dropdown is set to 'Current Positions' and the 'Index Allocation' dropdown is set to 'No Allocation'. The '\$delta cap' is set to '100,000,000'.

#Syms	symbol	acctnt	stk.status	alloc	delta	gamma	vega	pos.de	hedge	order	pos\$de
1	FB	T.SP1	TwoWay	None	-1,314	-27	-1,490	-1,314	1,314	1,310	-344,644
2	GME	T.SP1	TwoWay	None	90,212	0	0	90,212	-90,212	-17,773	10,168,690
3	NIO	T.SP1	TwoWay	None	23	0	-1	23	-23	0	1,159
4	NVDA	T.SP1	TwoWay	None	27	-1	-36	27	-27	-27	15,065
5	SPX	T.SP1	TwoWay	None	10,107	61	627,169	10,107	-10,107	-510	39,423,210
6	SPY	T.SP1	TwoWay	None	2,389	1	-14	2,389	-2,388	-2,388	930,390
7	TSLA	T.SP1	TwoWay	None	3,493	14	12,491	3,493	-3,493	-2,800	2,493,026
8	XLE	T.SP1	TwoWay	None	75	-11	-5	75	-74	0	3,729
9	ZOM	T.SP1	TwoWay	None	-50	0	0	-50	50	50	-105

If users wanted to allocate the deltas of a symbol position and hedge these deltas with a different symbol, they can select either 'Comp to Target: Uniform' or 'Comp to Target: Day Gm' to proxy hedge.

- 'Comp to Target: Uniform' will allocate the maximum of the symbol delta position or the \$ delta cap amount to the target symbol uniformly.
- 'Comp to Target: Day Gm' will allocate the maximum of the symbol delta position or the \$ delta cap amount to the target symbol gamma-weighted.



To set the Target symbol and Component symbol(s), users right-click on the 'alloc' cell of a specific symbol to select the appropriate value.

Below is an example where the user has chosen an Index Allocation of 'Component to Target: Uniform' setting SPX to **Component** and SPY to **Target**. By setting a \$delta cap of \$100,000 (see red arrow), the user is allocating a maximum of 100,000 \$deltas from SPX to SPY to hedge the SPX position.

SPIDERROCK Hedge Tool Quick filter: All Symbol(s): SPX, SPY All View: Regu

Hedge Once Start Loop Cancel Wave

Hedge Target: Current Positions Delta Index Allocation: Comp to Target: Uniform \$delta cap: 100,000 Bandwidth: hRatio: 1000.00 \$delta:

#Syms	pos\$delta	pos\$beta	opn\$delta	opn\$beta	shBot	shSld	bHdg\$De	sHdg\$De	Hdg\$De	bOrd\$De	sOrd\$De	Ord\$De
2	40.35 m	40.35 m	40.42 m	40.42 m	0	0	0	40.35 m	(40.35 m)	0	3.02 m	(3.02 m)

	symbol	acct	stk.status	alloc	delta	gamma	vega	pos.de	hedge	tgt\$de	order	alc\$de	pos\$de
1	SPX	T.SP1	TwoWay	Component	10,107	61	627,169	10,107	-10,081	39,325,240	-510	-100,000	39,423,210
2	SPY	T.SP1	TwoWay	Target	2,389	1	-14	2,389	-2,645	1,030,438	-2,645	100,000	930,390

Quick filter: All

Symbol(s): <ALL>

All

Vi

Bandwidth

hRatio: 0.15

\$delta: 250,000

z5Mult: 99

Portfolio

100 %

Reset

Max Order Size / Wave

shr: 300

\$de: 50,000,000

%: 100

☐ bal: 1.00

Execution Algorithm

algoType: AUTO.MID

tif: 10 s

sOrd\$De

51,655

Ord\$De

376,333

Sld	pos\$delta	tgt.ddelta	pos\$beta	gamma	vega	hRatio	alc.ddelta	hOpnDelta	pos.delta	trd.delta	upBand	dnBand	hedge	order
0	-248,289	-248,289	0	-725	-938	50.2	0	-23,259	-23,259	0	70	-70	23,189	0
0	-588,912	-588,912	-1,457,832	66	2,592	0.6	0	-507	-507	0	131	-131	375	300 1
0	-561,559	-561,559	0	7	-871	1.2	0	-212	-212	0	26	-26	185	0 2
0	10,502,430	10,502,430	15,341,050	7,330	5,771	4.6	0	60,995	60,995	0	1,452	-1,452	-59,543	-300
0	-273,723	-273,723	-273,723	7	-101	44.3	0	-1,030	-1,030	0	3	-3	1,026	300

• Bandwidth

The bandwidth values (hRatio, \$delta, z5Mult) enable users to define a risk tolerance. These values will create an up and down delta band to limit the hedging down to the nearest band. The number of shares needed to be bought/sold to hedge to the nearest band are displayed in the hedge column. Bandwidth values can be combined, but the tightest value will prevail to determine the up and down band.

• hRatio

Hedge Ratio (hRatio) is a value that represents the average 1-day move of the underlying price. ($\$deltas / \$gammas * \text{square root of } 252 \text{ days / implied vol}$)

Example: Assume your position in XYZ is 0 delta and 1000 gamma. If the calculated average daily move in the stock is 1.00 dollar per day, then the position will have swings of 1000 deltas on average. If the hedge ratio is set to "1.0" and the stock moves by 1.00 dollar, you'll have 1000 deltas and the hedge loop will not generate an order as the position is still within the 'bandwidth' of a 1.0 hedge ratio. If stock moves by \$1.50, the delta position will increase to 1500 and the hedge loop will generate an order for 500 shares to hedge the XYZ position back to the 1000 delta bandwidth.

• \$delta

This sets a risk tolerance measured in dollar deltas (underlyer price * delta).

• z5Mult

The Z5 bandwidth is a mathematical attempt to optimize the trade-off between the cost of hedging and the error in replicating a risk-free payoff (as in Black-Scholes original paper with 'continuous' rebalancing to replicate the payoff of a treasury). When a position's actual hedge exceeds a bandwidth around it's theoretical net delta, the position is re-hedged to the nearest bandwidth, not the theoretical delta (de).

Bandwidth

hRatio: 99.00 \$delta: 500,000 z5Mult: 99

Portfolio

90 % Reset

Max Order Size / Wave

shr: 600 \$de: 50,000,000 %: 100 ☐ bal: 1.00

Execution Algorithm

algoType: AUTO.MID tif: 10 s

sOrd\$De 1.05 m

Ord\$De (1.05 m)

Sld	pos\$delta	tgt.ddelta	pos\$beta	gamma	vega	hRatio	alc.ddelta	hOpnDelta	pos.delta	trd.delta	upBand	dnBand	hedge	order
0	-254,801	-254,801	0	-1,211	-946	99.0	0	-21,457	-21,457	0	42,105	-42,105	0	0
0	29,512,600	29,512,600	62,649,200	6,260	14,819	99.0	0	154,298	154,298	0	2,614	-2,614	-139,174	-600
0	1,173,605	1,173,605	2,928,381	80	904	99.0	0	998	998	0	425	-425	-570	-570
0	-454,825	-454,825	0	6	-1,376	99.0	0	-171	-171	0	188	-188	0	0
0	11,986,770	11,986,770	18,030,510	8,030	2,898	99.0	0	69,555	69,555	0	2,901	-2,901	-58,599	-600
0	2,400,047	2,400,047	2,400,047	12,045	2,689	99.0	0	9,001	9,001	0	1,875	-1,875	-6,899	-600

Portfolio

This box allows the user to set what percentage of their portfolio positions they would like to hedge.

Max Order Size / Wave

- shr: The shares box (shr) allows the user to set a maximum number of shares per order for each wave that the system is sending when hedging. In the example above, the user has set a maximum of 600 shares per order which is reflected in the order column. When the hedge target is below the number of shares, the system will create an order with a size corresponding to the hedge target.
- \$de: The \$delta box (\$de) allows users to set a maximum \$delta per wave sent to the system.
- %: This is the percentage of the portfolio to be hedged.
- bal: CCheking this box will balance the portfolio to be hedged based on a buy-to-sell ratio.
 - A bal ratio of 0.5 will balance twice as many sells as buys.
 - A bal ratio of 1 will balance buys and sells equally.
 - A balance ratio of 2 will balance twice as many buys as sells.

			Portfolio		Max Order Size / Wave									
\$delta:	0	z5Mult:	99	100 %	Reset	shr:	26,000	\$de:	2,000,000	%:	100	☒	bal:	0.50
shBot	shSld	bHdg\$De	sHdg\$De	Hdg\$De	bOrd\$De	sOrd\$De	Ord\$De							
0	0	136,048	1.44 m	(1.30 m)	99,762	198,610	(98,849)							
			Portfolio		Max Order Size / Wave									
\$delta:	0	z5Mult:	99	100 %	Reset	shr:	26,000	\$de:	2,000,000	%:	100	☐	bal:	1.00
shBot	shSld	bHdg\$De	sHdg\$De	Hdg\$De	bOrd\$De	sOrd\$De	Ord\$De							
0	0	136,048	1.44 m	(1.30 m)	99,762	1.32 m	(1.22 m)							

The purple boxes above display the current position \$delta (buy, sell and net) of the portfolio. The values in the green boxes show the anticipated \$delta impact before hedging. Modifying the bal ratio would change the values in those green boxes significantly as the hedging would lean differently based on this ratio.



• Execution Algorithm

- AlgoType: A variety of stock algos are available for users to work their orders more or less aggressively.
- SR.AUTO: Will take and make liquidity cycling from a passive to aggressive mode
- SR.TWAP: Same as above but will work in a TWAP fashion
- EAGLE, HAWK and FALCON will take liquidity when conditions are optimal based on the SpiderRock alpha probabilities. Eagle being the most passive and Falcon being the most aggressive.
- EAGLE.PRO, HAWK.PRO and FALCON.PRO work as above, but can post liquidity based on alpha probabilities in addition to taking liquidity.
- AUTO.CRX: Limit price goes out on the bid price if selling/on the ask price if buying
- AUTO.MID: Limit price goes out at mid-market and is restated every 5 seconds
- AUTO.TRN: Limit price turns the market by one price increment and is restated every 20 seconds

Execution Algorithm				
algoType:		AUTO.MID	tif: 10 sec	
		SR.AUTO		
		SR.TWAP		
		EAGLE		
d	dnBand	EAGLE.PRO	order	liveUPrc
5	-42,105	HAWK	0	11.88
4	-2,614	HAWK.PRO	-600	191.27
5	-425	FALCON	-570	1,176.38
8	-188	FALCON.PRO	0	2,662.08
1	-2,901	AUTO.CRX	-600	172.34
5	-1,875	AUTO.MID	-600	266.64
		AUTO.TRN		

- Tif: The time-in-force dropdown allows a user to select the amount of time a basket of orders will work, either once or in waves depending on whether the user has selected Hedge Once or Start Loop. The time frame ranges from 10 seconds to 30 minutes or extended day.



- **Quick filter**

Users can apply filters to their portfolio to isolate the positions with certain risk characteristics. The user can then hedge sections of their portfolio based on different risk tolerance criteria. The list of filters available is shown below. It is possible for users to have several instances of the hedge tool open so that they can create different risk tolerance levels for each sections of the portfolio determined by the filters. For example, symbols with a positive gamma might be hedged differently than symbols with a negative gamma.

- **Symbol(s)**

Users can also isolate one or several symbols using the Symbol(s) box. Symbols are delimited by a comma as shown below.

- **View**

The default view is Regular. Users can select Complex to see positions that derive from corporate actions.

Quick filter:		+gamma / +vega		Symbol(s): AMZN, NFLX		All		View: Regular	
		All		Execution Algorithm				Execution Risk	
10,000 %:		option trades		algoType: AUTO.MID				tif: 10 sec	
		naked stock position						\$delta: 50,000	
		earnings (-1 - 0)							
		earnings (0)							
Delta		pos.del		dnBand		hedge		order	
3,103		153,10		-2,619		-139,174		-600	
907		90		-425		-482		-482	
		+gamma / +vega							
		+gamma / -vega							
		-gamma / +vega							
		-gamma / -vega							
		pos (+) delta							
		neg (-) delta							
		pos (+) theta							
		neg (-) theta							
		pos (+) vega							
		neg (-) vega							
		pos (+) wvega							
		neg (-) wvega							
		pos (+) gamma							
		zero (0) gamma							
		neg (-) gamma							
		pos (+) theo.edge							
		neg (-) theo.edge							
		pos (+) open.pnl							
		neg (-) open.pnl							



HEDGING USING MULTIPLE ACCOUNTS

Selecting Accounts

Users are able to hedge from multiple accounts into one master account. In order to select the accounts to hedge from, manually type the accounts into the account box followed by a semi colon after each account, or select them from the account selector. Notice in the picture below, that the account 'T.CH' is in bold letters. This is the master account. All hedges will be done from combined positions from all accounts selected and the executions will take place in the master account.

The master account can be changed by double clicking on another account in the account selector or by placing an asterisk in front of the desired master account.

Symbol(s): <ALL> All View: Regular Test Account: *T.CH;T.CH2;T.CH3;T.CH4;T.CH5 Refresh

Filters: Clear All

Account	Name	Description	Client Firm	Client Account
T.CH	Chantelle Sampat Test		SR	TEST
T.CH2	Chantelle Sampat Beta		SR	TEST
T.CH3	Chantelle Sampat Te...		SR	TEST
T.CH4	Chantelle Debug		SR	
T.CH5	Chantelle Sampat Test		SR	TEST

The selected master account will be displayed in the acctn column as shown below.

SPIDERROCK Hedge Tool

Quick filter: All

Symbol(s): <ALL>

All

View: Regular

Test Account: *T.CH;T.CH2;T.CH3;T.CH4;T.CH5

Refresh

Hedge Once

Start Loop

Cancel Wave

Hedge Target

Open Positions

Delta

Index Allocation

Comp to Target: Uniform

\$delta cap: 75,000

Bandwidth

hRatio: 99.00

\$delta: 27,100

zMult: 99

Portfolio

100 %

Reset

Max Order Size / Wave

shr: 500,000

\$de: 50,000,000

%: 100

bat: 1.00

Execution Algorithm

algoType: AUTO.CRX

#Symb

pos\$delta

pos\$beta

opn\$delta

opn\$beta

shBot

shSld

bhdgsDe

shdgsDe

hdgsDe

bOrd\$De

sOrd\$De

Ord\$De

delta

posde

tgtd\$de

posbe

opt.trd.de

vega

hdg.ratid

z5.bwid

alc\$de

pos.de

hdg.opn.de

trd.de

56

677.70

570.53

573.88

685.14

57.88

0

331.86

700.88

369.02

140.03

421.18

281.15

0

12,215,190

12,215,190

12,208,470

14,418,860

0

0

99.0

0

448,264

448,264

0

1

Z

*T.CH

TwoWay

None

448,264

IV

14,418,860

0

0

0

0

0

0

0

0

12,215,190

12,215,190

12,208,470

14,418,860

0

0

99.0

0

448,264

448,264

0

2

XLY

*T.CH

TwoWay

None

10,261

IV

1,036,980

0

0

0

0

0

0

0

0

1,069,870

1,069,870

1,069,248

1,036,980

0

0

99.0

0

10,261

10,261

0

3

XLIF

*T.CH

TwoWay

None

1,894

IV

64,198

0

0

0

0

0

0

0

0

50,599

50,599

50,579

64,198

0

0

99.0

0

1,894

1,894

0

4

XLE

*T.CH

TwoWay

None

854,724

IV

50,793,190

0

0

0

0

0

0

0

57,270,800

57,270,800

57,253,690

50,793,190

0

0

99.0

0

854,724

854,724

0

5

X

*T.CH

TwoWay

None

1,026,784

IV

66,706,690

0

-263

-1,952

0

0

0

0

-27,974,690

-27,974,690

-27,974,730

66,706,690

0

-402

99.0

-210

0

1,026,784

1,026,784

0

6

WBA

*T.CH

TwoWay

None

293,099

IV

19,766,370

0

224

15,146

0

0

0

0

24,104,420

24,104,420

24,091,250

19,766,370

0

2,605

99.0

594

0

293,099

293,099

0

7

WAL

*T.CH

TwoWay

None

3,418

IV

224,731

0

135

2,942

0

0

0

0

159,772

159,772

159,739

224,731

0

804

99.0

-128

0

3,418

3,418

0

8

W

*T.CH

TwoWay

None

716,505

IV

74,038,940

0

0

0

0

0

0

0

60,924,410

60,924,410

60,752,460

74,038,940

0

0

99.0

0

716,505

716,505

0

9

VOX

*T.CH

TwoWay

None

171,365

IV

-31,016,980

0

77

1,031

0

0

0

0

6,288,276

6,288,276

6,296,024

-31,016,980

0

24

99.0

67

0

171,365

171,365

0

10

VIX

*T.CH

TwoWay

None

-62

IV

-1,205

0

-17

-66

0

0

0

0

-1,205

-1,205

-1,189

-1,205

0

-1

99.0

-8

0

-62

-62

0

11

V

*T.CH

TwoWay

None

647,762

IV

85,601,400

0

0

0

0

0

0

0

87,285,930

87,285,930

87,260,020

85,601,400

0

0

99.0

0

647,762

647,762

0

12

UPS

*T.CH

TwoWay

None

155

IV

22,493

0

0

0

0

0

0

0

17,239

17,239

17,233

22,493

0

0

99.0

0

155

155

0

13

TWTR

*T.CH

TwoWay

None

208

IV

2,541

0

0

0

0

0

0

0

6,676

6,676

6,670

2,541

0

0

99.0

0

208

208

0

14

TSM

*T.CH

TwoWay

None

-49,800

IV

-1,801,629

0

4,569

60,251

0

 0 | 0 | 0 | -1,808,498 | -1,808,498 | -1,807,999 | -1,801,629 | 0 | 1,310 | 99.0 | 4,802 | 0 | -49,800 | -49,800 | 0 |

15

TSLA

*T.CH

TwoWay

None

55,683

IV

9,983,754

0

1,449

1,833,780

340

 0 | 0 | 0 | 19,811,350 | 19,811,350 | 27,895,650 | 9,983,754 | -22,213 | 107,074 | 99.0 | 6,571 | 0 | 55,683 | 78,464 | -22,213 |

16

TRLY

*T.CH

TwoWay

None

22,985

IV

2,398,230

0

4

461

 0 | 0 | 0 | 2,398,230 | 2,398,230 | 2,401,555 | 2,398,230 | 0 | 15 | 99.0 | 12 | 0 | 22,985 | 22,985 | 0 |

17

TDX

*T.CH

TwoWay

None

-949,259

IV

-23,606,480

0

 0 | 0 | 0 | 0 | 0 | 0 | -46,561,100 | -46,561,100 | -46,551,660 | -23,606,480 | 0 | 0 | 99.0 | 1,111 | 0 | -949,259 | -949,259 | 0 |

18

TIF

*T.CH

TwoWay

None

320,273

IV

24,198,280

0 0 | 0 | 0 | 0 | 0 | 32,890,480 | 32,890,480 | 32,880,830 | 24,198,280 | 0 | 0 | 99.0 | 0 | 320,273 | 320,273 | 0 |

19

T

*T.CH

TwoWay

None

-431,550

IV

114,213,800

0

 1,310 | 12,030 | 30 | 0 | 0 | 134,298,200 | 134,298,200 | 134,338,500 | 114,213,800 | -1,215 | 68 | 99.0 | 400 | 0 | -431,550 | -431,550 | -1,215 |

20

ST

*T.CH

TwoWay

None

-4,597

IV

-310,232

0

 -756 | -16,368 | 0 | 0 | 0 | -213,863 | -213,863 | -213,794 | -310,232 | 0 | -427 | 99.0 | -1,177 | 0 | -4,597 | -4,597 | 0 |

21

SPY

*T.CH

TwoWay

Target

97,119

IV

26,152,050

0

 6,262 | 4,540,652 | 0 | 0 | 0 | 26,152,050 | 26,151,580 | 25,929,370 | 26,151,580 | 0 | 74,482 | 99.0 | 16,776 | -75,000 | 97,117 | 96,617 | 0 |

22

SPX

*T.CH

TwoWay

Component

-69,273

IV

-187,108,500

0

 578 | -41,899,940 | 200 | 0 | 0 | -186,475,400 | -189,588,000 | -189,432,600 | -190,231,700 | -446 | 361,124 | 99.0 | 4,460 | 75,000 | -70,429 | -70,429 | -446 |

23

SODA

*T.CH

TwoWay

None

123

IV

23,864 0 | 0 | 0 | 0 | 0 | 0 | 17,627 | 17,627 | 17,624 | 23,864 | 0 | 0 | 99.0 | 0 | 123 | 123 | 0 |

24

SNAP

*T.CH

TwoWay

None

17,910

IV

15,192 0 | 0 | 0 | 0 | 0 | 0 | 109,163 | 109,163 | 109,162 | 15,192 | 0 | 0 | 99.0 | 0 | 17,910 | 17,910 | 0 |

25

SHAK

*T.CH

TwoWay

None

46,515

IV

2,180,858 0 | 0 | 0 | 0 | 0 | 0 | 2,337,616 | 2,337,616 | 2,336,914 | 2,180,858 | 0 | 0 | 99.0 | 0 | 46,515 | 46,515 | 0 |

